2.2 Complex Functions as Mappings

INTRODUCTION Recall that if f is a real-valued function of a real variable, then the graph of f is a curve in the Cartesian plane. Graphs are used extensively to investigate properties of *real* functions in elementary courses. However, we'll see that the graph of a *complex* function lies in four-dimensional space, and so we cannot use graphs to study complex functions. In this section we discuss the concept of a *complex mapping*, which was developed by the German mathematician Bernhard Riemann to give a geometric representation of a complex function. The basic idea is this. Every complex function describes a correspondence between points in two copies of the complex plane. Specifically, the point z in the z-plane is associated with the unique point $w = f(z)$ in the w-plane. We use the alternative term **complex mapping** in place of "complex function" when considering the function as this correspondence between points in the z-plane and points in the w-plane. The geometric representation of a complex mapping $w = f(z)$ consists of two figures: the first, a subset S of points in the z-plane, and the second, the set S' of the images of points in S under $w = f(z)$ in the w-plane.

Mappings

A useful tool for the study of real functions in elementary calculus is the graph of the function. Recall that if $y = f(x)$ is a real-valued function of a real variable x, then the graph of f is defined to be the set of all points $(x, f(x))$ in the two-dimensional Cartesian plane. An analogous definition can be made for a complex function. However, if $w = f(z)$ is a complex function, then both z and w lie in a complex plane. It follows that the set of all points $(z, f(z))$ lies in four-dimensional space (two dimensions from the input z and two dimensions from the output w). Of course, a subset of four-dimensional space cannot be easily illustrated. Therefore:

- *We cannot draw the graph of a complex function.*

The concept of a complex mapping provides an alternative way of giving a geometric representation of a complex function. As described in the section introduction, we use the term **complex mapping** to refer to the correspondence determined by a complex function $w = f(z)$ between points in a z-plane and images in a w-plane. If the point z_0 in the z-plane corresponds to the point w_0 in the w-plane, that is, if $w_0 = f(z_0)$, then we say that f **maps** z_0 onto w_0 or, equivalently, that z_0 is **mapped** onto w_0 by f.

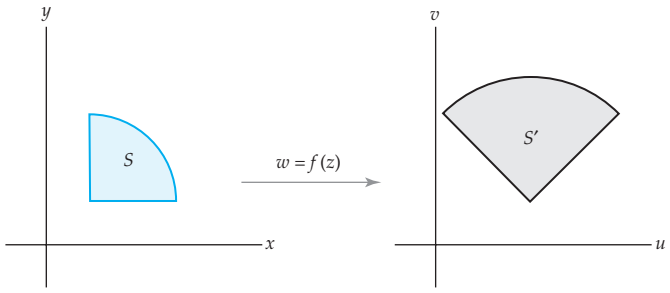
As an example of this type of geometric thinking, consider the real function $f(x) = x + 2$. Rather than representing this function with a line of slope 1 and y-intercept (0, 2), consider how one copy of the real line (the x-line) is mapped onto another copy of the real line (the y-line) by f. Each point on the x-line is mapped onto a point two units to the right on the y-line (0 is mapped onto 2, while 3 is mapped onto 5, and so on). Therefore, the real function $f(x) = x + 2$ can be thought of as a mapping that *translates* each point in the real line two units to the right. You can visualize the action of this mapping by imagining the real line as an infinite rigid rod that is physically moved two units to the right.

In order to create a geometric representation of a complex mapping, we begin with two copies of the complex plane, the z-plane and the w-plane, drawn either side-by-side or one above the other. A complex mapping is represented by drawing a set S of points in the z-plane and the corresponding set of images of the points in S under f in the w-plane. This idea is illustrated in **FIGURE 2.2.1** where a set S in the z-plane is shown in color in Figure 2.2.1(a) and a set labeled S', which represents the set of the images of points in S under $w = f(z)$, is shown in gray in Figure 2.2.1(b). From this point on we will use notation similar to that in Figure 2.2.1 when discussing mappings.

Notation: S'

If $w = f(z)$ is a complex mapping and if S is a set of points in the z-plane, then we call the set of images of the points in S under f the ***image of S under f****, and we denote this set by the symbol S'.**

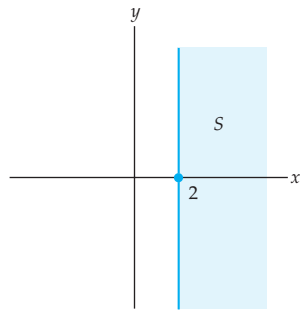
If the set S has additional properties, such as S is a domain or a curve, then we also use symbols such as D and D' or C and C', respectively, to denote the set and its image under a complex mapping. The notation $f(C)$ is also sometimes used to denote the image of a curve C under $w = f(z)$.



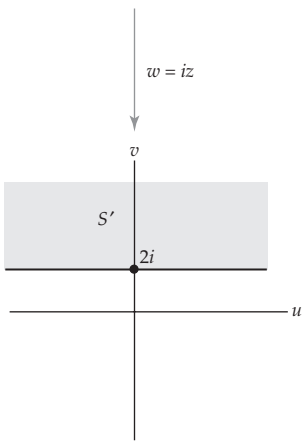
(a) The set S in the z-plane **(b)** The image of S in the w-plane

FIGURE 2.2.1 The image of a set S under a mapping $w = f(z)$

An illustration like Figure 2.2.1 is meant to convey information about the general relationship between an arbitrary point z and its image $w = f(z)$. As such, the set S needs to be chosen with some care. For example, if f is a function whose domain and range are the set of complex numbers $\mathbf{C}$, then choosing $S = \mathbf{C}$ will result in a figure consisting solely of two complex planes. Clearly, such an illustration would give no insight into *how* points in the z-plane are mapped onto points in the w-plane by f.



(a) The half-plane S



(b) The image, S', of the half-plane S

FIGURE 2.2.2 The mapping $w = iz$

EXAMPLE 1 Image of a Half-Plane under $w = iz$

Find the image of the half-plane $\text{Re}(z) \geq 2$ under the complex mapping $w = iz$ and represent the mapping graphically.

Solution Let S be the half-plane consisting of all complex points z with $\text{Re}(z) \geq 2$ shown in color in FIGURE 2.2.2(a). This set can be described by the two simultaneous inequalities,

$$x \geq 2 \quad \text{and} \quad -\infty < y < \infty. \tag{1}$$

In order to describe the image of S, we express the mapping $w = iz$ in terms of its real and imaginary parts u and v; then we use the bounds given by (1) on x and y in the z-plane to determine bounds on u and v in the w-plane. By replacing the symbol z by $x + iy$ in $w = iz$, we obtain $w = i(x + iy) = -y + ix$, and so the real and imaginary parts of $w = iz$ are:

$$u = -y \quad \text{and} \quad v = x. \tag{2}$$

After replacing the variables x and y in (1) using the equations $x = v$ and $y = -u$ from (2), we obtain $v \geq 2$ and $-\infty < u < \infty$. That is, the set S', the image of S under $w = iz$, consists of all points $w = u + iv$ in the w-plane that satisfy the

*The set S is sometimes called the **pre-image** of S' under f.

simultaneous inequalities $v \geq 2$ and $-\infty < u < \infty$. In words, the set S' consists of all points in the half-plane lying on or above the horizontal line $v = 2$. This image can also be described by the single inequality $\text{Im}(w) \geq 2$. In summary, the half-plane $\text{Re}(z) \geq 2$ shown in color in Figure 2.2.2(a) is mapped onto the half-plane $\text{Im}(w) \geq 2$ shown in gray in Figure 2.2.2(b) by the complex mapping $w = iz$.

In Example 1, the set S and its image S' are both half-planes. This might lead you to believe that there is some simple geometric way to visualize the image of other sets in the complex plane under the mapping $w = iz$. (We will see that this is the case in Section 2.3.) For most mappings, however, the relationship between S and S' is more complicated. This is illustrated in the following example.

EXAMPLE 2 Image of a Line under $w = z^2$

Find the image of the vertical line $x = 1$ under the complex mapping $w = z^2$ and represent the mapping graphically.

Solution Let C be the set of points on the vertical line $x = 1$ or, equivalently, the set of points $z = 1 + iy$ with $-\infty < y < \infty$. We proceed as in Example 1. From (1) of Section 2.1, the real and imaginary parts of $w = z^2$ are $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$, respectively. For a point $z = 1 + iy$ in C, we have $u(1, y) = 1 - y^2$ and $v(1, y) = 2y$. This implies that the image of S is the set of points $w = u + iv$ satisfying the simultaneous equations:

$$u = 1 - y^2 \tag{3}$$

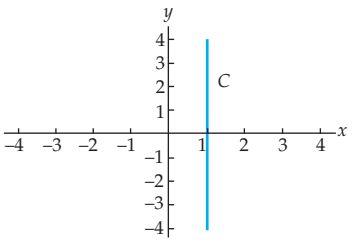
and

$$v = 2y \tag{4}$$

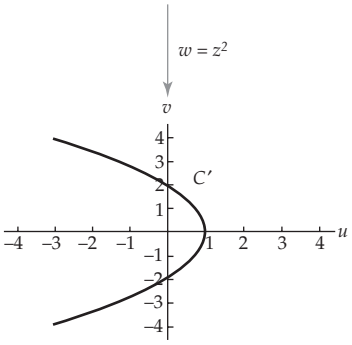
for $-\infty < y < \infty$. Equations (3) and (4) are *parametric equations* in the real parameter y, and they define a curve in the w-plane. We can find a Cartesian equation in u and v for this curve by eliminating the parameter y. In order to do so, we solve (4) for y and then substitute this expression into (3):

$$u = 1 - \left(\frac{v}{2}\right)^2 = 1 - \frac{v^2}{4}. \tag{5}$$

Since y can take on any real value and since $v = 2y$, it follows that v can take on any real value in (5). Consequently, C'—the image of C—is a parabola in the w-plane with vertex at (1,0) and u-intercepts at $(0, \pm 2)$. See FIGURE 2.2.3(b). In conclusion, we have shown that the vertical line $x = 1$ shown in color in Figure 2.2.3(a) is mapped onto the parabola $u = 1 - \frac{1}{4}v^2$ shown in black in Figure 2.2.3(b) by the complex mapping $w = z^2$.



(a) The vertical line $\text{Re}(z) = 1$



(b) The image of C is the parabola $u = 1 - \frac{1}{4}v^2$

FIGURE 2.2.3 The mapping $w = z^2$

In contrast to Figure 2.2.2, the representation of the mapping $w = z^2$ shown in Figure 2.2.3 gives little insight into what the images of other sets in the plane might be. Mapping by this "complex squaring function" will be examined in greater detail in Section 2.4.

Parametric Curves in the Complex Plane

For a simple complex function, the manner in which the complex plane is mapped might be evident after analyzing the image of a single set, but for most functions an understanding of the mapping is obtained only after looking at the images of a variety of sets. We can often gain a good understanding of a complex mapping by analyzing the

images of *curves* (one-dimensional subsets of the complex plane) and this process is facilitated by the use of parametric equations.

A review of parametric curves can be found in Section 2.2 of the student study guide. ▶

If $x = x(t)$ and $y = y(t)$ are real-valued functions of a real variable t, then the set C of all points $(x(t),\ y(t))$, where $a \le t \le b$, is called a **parametric curve**. The equations $x = x(t)$, $y = y(t)$, for $a \le t \le b$ are called **parametric equations** of C. A parametric curve can be regarded as lying in the complex plane by letting x and y represent the real and imaginary parts of a point in the complex plane. In other words, if $x = x(t)$, $y = y(t)$, and $a \le t \le b$ are parametric equations of a curve C in the Cartesian plane, then the equation $z(t) = x(t) + iy(t)$, $a \le t \le b$, is a description of the curve C in the complex plane. For example, consider the parametric equations $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$, of a curve C in the xy-plane (the curve C is a circle centered at $(0, 0)$ with radius 1). The equation $z(t) = \cos t + i \sin t$, $0 \le t \le 2\pi$, describes the same curve C in the complex plane. If, say, $t = \pi/2$, then the point $(\cos \frac{\pi}{2}, \sin \frac{\pi}{2}) = (0, 1)$ is on the curve C in the Cartesian plane, while the point $z(\pi/2) = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i$ represents this point on C in the complex plane. This discussion is summarized in the following definition.

In a parametrization $z(t) = x(t) + iy(t)$, t is a **real** variable. ▶

DEFINITION 2.2.1 Parametric Curves in the Complex Plane

If $x(t)$ and $y(t)$ are real-valued functions of a real variable t, then the set C consisting of all points $z(t) = x(t) + iy(t)$, $a \le t \le b$, is called a **parametric curve** or a **complex parametric curve**. The complex-valued function of the real variable t, $z(t) = x(t) + iy(t)$, is called a **parametrization** of C.

Properties of curves in the Cartesian plane such as continuous, differentiable, smooth, simple, and closed can all be reformulated to be properties of curves in the complex plane. These properties are important in the study of the complex integral and will be discussed in Chapter 5.

Two of the most elementary curves in the plane are lines and circles. Parametrizations of these curves in the complex plane can be derived from parametrizations in the Cartesian plane. It is also relatively easy to find these parametrizations directly by using the geometry of the complex plane. For example, suppose that we wish to find a parametrization of the line in the complex plane containing the points z_0 and z_1. We know from Chapter 1 that $z_1 - z_0$ represents the vector originating at z_0 and terminating at z_1, shown in color in FIGURE 2.2.4. If z is any point on the line containing z_0 and z_1, then inspection of Figure 2.2.4 indicates that the vector $z - z_0$ is a real multiple of the vector $z_1 - z_0$. Therefore, if z is on the line containing z_0 and z_1, then there is a real number t such that $z - z_0 = t\,(z_1 - z_0)$. Solving this equation for z gives a parametrization $z(t) = z_0 + t\,(z_1 - z_0) = z_0\,(1 - t) + z_1 t$, $-\infty < t < \infty$, for the line. Note that if we restrict the parameter t to the interval $[0, 1]$, then the points $z(t)$ range from z_0 to z_1, and this gives a parametrization of the line segment from z_0 to z_1. On the other hand, if we restrict t to the interval $[0, \infty]$, then we obtain a parametrization of the ray emanating from z_0 and containing z_1. These parametrizations are included in the following summary.

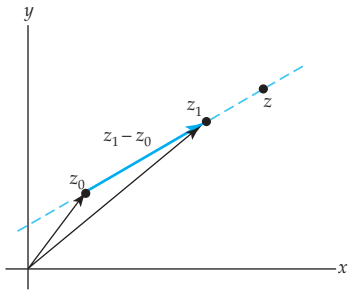


FIGURE 2.2.4 Parametrization of a line

Common Parametric Curves in the Complex Plane

Line

A parametrization of the line containing the points z_0 and z_1 is:

These are not the only parametrizations possible! See Problem 28 in Exercises 2.2. ▶

$$z(t) = z_0(1 - t) + z_1 t, \quad -\infty < t < \infty. \tag{6}$$

Line Segment
A parametrization of the line segment from z_0 to z_1 is:

$$z(t) = z_0(1-t) + z_1 t, \quad 0 \le t \le 1. \tag{7}$$

Ray
A parametrization of the ray emanating from z_0 and containing z_1 is:

$$z(t) = z_0(1-t) + z_1 t, \quad 0 \le t < \infty. \tag{8}$$

Circle
A parametrization of the circle centered at z_0 with radius r is:

$$z(t) = z_0 + r(\cos t + i \sin t), \quad 0 \le t \le 2\pi. \tag{9}$$

In exponential notation, this parametrization is:

$$z(t) = z_0 + re^{it}, \quad 0 \le t \le 2\pi. \tag{10}$$

Parametric curves are important in the study of complex mappings because it is easy to determine a parametrization of the image of a parametric curve. For example, if $w = iz$ and C is the line $x = 2$ given by $z(t) = 2 + it$, $-\infty < t < \infty$, then the value of $f(z) = iz$ at a point on this line is $w = f(2+it) = i(2+it) = -t + 2i$, and so the image of $z(t)$ is $w(t) = -t + 2i$. Put another way, $w(t) = -t + 2i$, $-\infty < t < \infty$, is a parametrization of the image C'. Thus, C' is the line $v = 2$. In summary, we have the following procedure for finding the images of curves under a complex mapping.

Image of a Parametric Curve under a Complex Mapping

If $w = f(z)$ is a complex mapping and if C is a curve parametrized by $z(t)$, $a \le t \le b$, then

$$w(t) = f(z(t)), \quad a \le t \le b \tag{11}$$

is a parametrization of the image, C' of C under $w = f(z)$.

In some instances it is convenient to represent a complex mapping using a single copy of the complex plane. We do so by superimposing the w-plane on top of the z-plane, so that the real and imaginary axes in each copy of the plane coincide. Because such a figure simultaneously represents both the z- and the w-planes, we omit all labels x, y, u, and v from the axes. In the following examples, we represent a complex mapping using a single copy of the complex plane.

EXAMPLE 3 Image of a Parametric Curve

Use (11) to find the image of the line segment from 1 to i under the complex mapping $w = \overline{iz}$.

Solution Let C denote the line segment from 1 to i and let C' denote its image under $f(z) = \overline{iz}$. By identifying $z_0 = 1$ and $z_1 = i$ in (7), we obtain a parametrization $z(t) = 1 - t + it$, $0 \le t \le 1$, of C. The image C' is then given by (11):

$$w(t) = f(z(t)) = \overline{i(1-t+it)} = -i(1-t) - t, \quad 0 \le t \le 1.$$

With the identifications $z_0 = -i$ and $z_1 = -1$ in (7), we see that $w(t)$ is a parametrization of the line segment from $-i$ to -1. Therefore, C' is the line segment from

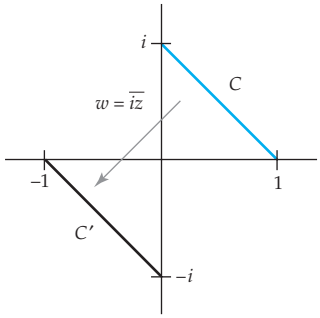


FIGURE 2.2.5 The mapping $w = \overline{iz}$

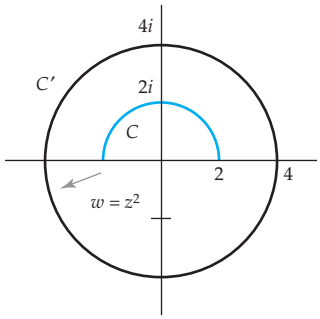


FIGURE 2.2.6 The mapping $w = z^2$

$-i$ to -1. This mapping is depicted in FIGURE 2.2.5 using a single copy of the complex plane. In Figure 2.2.5, the line segment shown in color is mapped onto the line segment shown in black by $w = \overline{iz}$.

EXAMPLE 4 Image of a Parametric Curve

Find the image of the semicircle shown in color in FIGURE 2.2.6 under the complex mapping $w = z^2$.

Solution Let C denote the semicircle shown in Figure 2.2.6 and let C' denote its image under $f(z) = z^2$. By setting $z_0 = 0$ and $r = 2$ in (10) we obtain the following parametrization of C:

$$z(t) = 2e^{it}, \quad 0 \le t \le \pi.$$

Thus, from (11) we have that:

$$w(t) = f(z(t)) = \left(2e^{it}\right)^2 = 4e^{2it}, \quad 0 \le t \le \pi, \tag{12}$$

is a parametrization of C'. If we set $t = \frac{1}{2}s$ in (12), then we obtain a new parametrization of C':

$$W(s) = 4e^{is}, \ 0 \le s \le 2\pi. \tag{13}$$

From (10) with $z_0 = 0$ and $r = 4$, we find that (13) defines a circle centered at 0 with radius 4. Therefore, the image C' is the circle $|w| = 4$. We represent this mapping in Figure 2.2.6 where the semicircle shown in color is mapped onto the circle shown in black by $w = z^2$.

Use of Computers

Computer algebra systems such as *Maple* and *Mathematica* perform standard algebraic operations with complex numbers. This capability combined with the ability to graph a parametric curve makes these systems excellent tools for exploring properties of complex mappings. In *Mathematica*, for example, a complex function can be defined using the command

f[z_] := *an expression in z.*

A complex parametrization can be defined similarly using the command

g[t_] := *an expression in t.*

From (11), it follows that **w[t_] := f[g[t]]** is a parametrization of the image of the curve. This image can be graphed using the parametric plot command:

ParametricPlot[{**Re**[**w**[**t**]], **Im**[**w**[**t**]]}, {**t**, **a**, **b**}]

where **a** and **b** are the upper and lower bounds on **t** respectively. For example, *Mathematica* was used to produce FIGURE 2.2.7, which shows the image of the circle $|z| = 2$ under the complex mapping $w = z^2 + iz - \text{Re}(z)$.

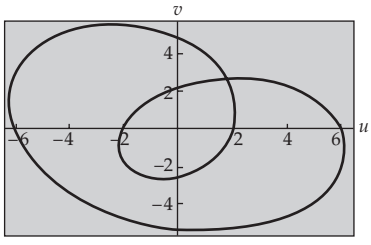


FIGURE 2.2.7 The image of a circle under $w = z^2 + iz - \text{Re}(z)$

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) In this section we introduced an important difference between real and complex analysis, namely, that we cannot graph a complex function. Instead, we represent a complex function with two figures: the first a subset S in the complex plane, and the second, the image S' of the set S under a complex mapping. A complete understanding of a complex mapping is obtained when we understand the relationship between *any* set S and its image S'.

(*ii*) Complex mappings are closely related to parametric curves in the plane. In later sections, we use this relationship to help visualize the notions of limit, continuity, and differentiability of complex functions. Parametric curves will also be of central importance in the study of complex integrals much as they were in the study of real line integrals.

EXERCISES 2.2

Answers to selected odd-numbered problems begin on page ANS-6.

In Problems 1–8, proceed as in Example 1 or Example 2 to find the image S' of the set S under the given complex mapping $w = f(z)$.

1. $f(z) = i\bar{z}$; S is the horizontal line $y = 3$

2. $f(z) = \bar{z}$; S is the line $y = x$

3. $f(z) = 3z$; S is the half-plane $\text{Im}(z) > 2$

4. $f(z) = 3iz$; S is the infinite vertical strip $2 \le \text{Re}(z) < 3$

5. $f(z) = (1 + i)z$; S is the vertical line $x = 2$

6. $f(z) = (1 - i)z$; S is the line $y = 2x + 1$

7. $f(z) = iz + 4$; S is the half-plane $\text{Im}(z) \le 1$

8. $f(z) = iz + \bar{z}$; S is the vertical line $x = 3$

In Problems 9–14, find the image of the given line under the complex mapping $w = z^2$.

9. $y = 1$

10. $x = -3$

11. $x = 0$

12. $y = 0$

13. $y = x$

14. $y = -x$

In Problems 15–20, (**a**) plot the parametric curve C given by $z(t)$ and describe the curve in words, (**b**) find a parametrization of the image, C', of C under the given complex mapping $w = f(z)$, and (**c**) plot C' and describe this curve in words.

15. $z(t) = 2(1 - t) + it, 0 \le t \le 1;\ f(z) = 3z$

16. $z(t) = i(1 - t) + (1 + i)t,\ 0 \le t < \infty;\ f(z) = -z$

17. $z(t) = 1 + 2e^{it},\ 0 \le t \le 2\pi;\ f(z) = z + 1 - i$

18. $z(t) = i + e^{it},\ 0 \le t \le \pi;\ f(z) = (z - i)^2$

19. $z(t) = t,\ 0 \le t \le 2;\ f(z) = e^{i\pi z}$

20. $z(t) = 4e^{it},\ 0 \le t \le \pi,\ f(z) = \text{Re}(z)$

In Problems 21–26, use parametrizations to find the image, C', of the curve C under the given complex mapping $w = f(z)$.

21. $f(z) = z^3$; C is the positive imaginary axis

22. $f(z) = iz$; C is the circle $|z - 1| = 2$

23. $f(z) = 1/z$; C is the circle $|z| = 2$

24. $f(z) = 1/z$; C is the line segment from $1 - i$ to $2 - 2i$

25. $f(z) = z + \bar{z}$; C is the semicircle of the unit circle $|z| = 1$ in the upper half-plane $\text{Im}(z) \geq 0$

26. $f(z) = e^z$; C is the ray emanating from the origin and containing $2 + \sqrt{3}i$

Focus on Concepts

27. In this problem we will find the image of the line $x = 1$ under the complex mapping $w = 1/z$.

(**a**) The line $x = 1$ consists of all points $z = 1 + iy$ where $-\infty < y < \infty$. Find the real and imaginary parts u and v of $f(z) = 1/z$ at a point $z = 1 + iy$ on this line.

(**b**) Show that $\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}$ for the functions u and v from part (a).

(**c**) Based on part (b), describe the image of the line $x = 1$ under the complex mapping $w = 1/z$.

(**d**) Is there a point on the line $x = 1$ that maps onto 0? Do you want to alter your description of the image in part (c)?

28. Consider the parametrization $z(t) = i(1 - t) + 3t$, $0 \leq t \leq 1$.

(**a**) Describe in words this parametric curve.

(**b**) What is the difference between the curve in part (a) and the curve defined by the parametrization $z(t) = 3(1 - t) + it$, $0 \leq t \leq 1$?

(**c**) What is the difference between the curve in part (a) and the curve defined by the parametrization $z(t) = \frac{3}{2}t + i\left(1 - \frac{1}{2}t\right)$, $0 \leq t \leq 2$?

(**d**) Find a parametrization of the line segment from $1 + 2i$ to $2 + i$ where the parameter satisfies $0 \leq t \leq 3$.

29. Use parametrizations to find the image of the circle $|z - z_0| = R$ under the mapping $f(z) = iz - 2$. Describe the image in words.

30. Consider the line $y = mx + b$ in the complex plane.

(**a**) Give a parametrization $z(t)$ for the line.

(**b**) Describe in words the image of the line under the complex mapping $w = z + 2 - 3i$.

(**c**) Describe in words the image of the line under the complex mapping $w = 3z$.

31. The complex mapping $w = \bar{z}$ is called **reflection** about the real axis. Explain why.

32. Let $f(z) = az$ where a is a complex constant and $|a| = 1$.

(**a**) Show that $|f(z_1) - f(z_2)| = |z_1 - z_2|$ for all complex numbers z_1 and z_2.

(**b**) Give a geometric interpretation of the result in (a).

(**c**) What does your answer to (b) tell you about the image of a circle under the complex mapping $w = az$.

33. In this problem we investigate the effect of the mapping $w = az$, where a is a complex constant and $a \neq 0$, on angles between rays emanating from the origin.

(**a**) Let C be a ray in the complex plane emanating from the origin. Use parametrizations to show that the image C' of C under $w = az$ is also a ray emanating from the origin.